01 — Project Proposal

Project name

SeatMe — a serverless event guest-list and seating manager.

Problem

Planning the guest list and seating for an event (wedding, conference, party) is usually handled with spreadsheets and group chats. This is error-prone: RSVPs are scattered, party sizes are forgotten, table capacities are exceeded, and there is no single place that shows who is coming and where they sit. Organizers also have no easy way to collect responses from guests who do not want to install an app or create an account.

Proposed solution

SeatMe is a cloud, fully serverless web application with two audiences:

- **Hosts** create an event, manage a categorized guest list, define tables and capacities, automatically generate a seating plan, and send each guest a personal RSVP link by email.
- **Guests** open a personal link (no account, no login) and reply *yes/no*, set their party size, and optionally request a song. Their response appears instantly on the host's dashboard.

A **platform administrator** can view and manage every event on the platform.

Target users / permission groups

1. **User (host)** — an authenticated event organizer who manages their own event.
2. **Admin** — a platform administrator who oversees and can manage all events.
3. **Guest** — an unauthenticated invitee who only uses their personal RSVP link.

Why serverless / cloud

- **Scales to zero** when idle and scales out automatically for bursts (e.g. when invitations go out and many guests RSVP at once).
- **No servers to patch or operate** — every component is an AWS managed service.
- **Pay-per-use** — cost is dominated by actual requests, which suits spiky, event-driven traffic.

Core AWS building blocks

Concern	Service
Web hosting	Amazon S3 static website
Identity & auth	Amazon Cognito User Pool
API	Amazon API Gateway (HTTP API)
Business logic	AWS Lambda (Python 3.12)

Concern	Service
Data	Amazon DynamoDB
Notifications	Amazon SNS

Scope (MVP delivered)

- Host sign-up / sign-in / password reset (Cognito).
- Create / view / edit / delete an event.
- Manage categories, tables and capacities.
- Add / edit / delete guests; manual seat assignment.
- Public RSVP page reachable from a personal link.
- Automatic, category-aware seating generation.
- Email invitations (single or bulk) via SNS.
- Admin portal listing and managing all events.

Out of scope (future work)

- Attaching a Cognito JWT authorizer to the HTTP API (write endpoints and the admin route are currently authorized inside their Lambdas; public reads remain open for RSVP).
- Drag-and-drop seating UI.
- Multiple events per host.

Success criteria

- A host can go from sign-up to a generated seating plan and sent invitations without leaving the browser.
- A guest can RSVP from a link with no account.
- An admin can see every event and open/delete any of them.
- The entire stack deploys from a single command (`python redeploy_all.py`).